

05 — Complete Project Execution Flow

This document details the exact sequence of events that occurs from the moment a user launches the browser application to when machine learning predictions and interactive charts update on screen.

■ End-to-End User Interaction Flow Diagram

User Opens Browser (`http://localhost:8501`) ■ ▼ Streamlit App Execution (``frontend/app.py``) ■ ▼ Renders Header & Top Hero Objective Banner ■ ▼ Tab 3 Activated ("■ Cargo Forecasting") ■ ▼ API Call: ``api_get("/cargo/commodities")`` ■ ▼ FastAPI Router (``backend/app/routes/cargo.py:get_commodities()``) ■ ▼ Service Call (``backend/app/services/forecasting.py:get_available_commodities()``) ■ ▼ Returns List of 16+ Commodities to Frontend Dropdown ■ ▼ User Selects Commodity: "TOTAL COAL", Horizon: "6 Months" ■ ▼ API Call: ``api_get("/cargo/forecast", {"horizon": 6, "commodity": "TOTAL COAL", "section": "ALL"})`` ■ ▼ FastAPI Router (``backend/app/routes/cargo.py:get_forecast()``) ■ ▼ Service Call (``backend/app/services/forecasting.py:get_enhanced_cargo_forecast()``) ■ ▼ Data Ingestion & Preprocessing (``_load_and_prep_cargo_data()``) ■ ▼ Holt-Winters & Ridge ML Regression Execution ■ ▼ Computes 95% Prediction Interval Bands & WAPE Metrics ■ ▼ Calculates Driver Weights (Monsoon, Hinterland demand, UPCL Power) ■ ▼ API Returns JSON Response Payload (HTTP 200 OK) ■ ▼ Frontend Receives Response (``frontend/app.py``) ■ ▼ UI Updates Dynamically: • Selected Commodity Metrics Banner • NMPA Facility Mapping (Berths 15 & 16 UPCL) • Plotly Line Chart (Historical Cargo vs Predicted Cargo vs 95% Band) • Explainable Forecast Drivers Cards • Model Validation Backtesting Table (WAPE vs Baseline)

■■ Step-by-Step Code Execution Trace

Step 1: User Launches Application

- **Action:** User opens browser to *CODE_PH_0*.
- **Code Execution:** Streamlit executes *CODE_PH_0* top-to-bottom.
- **UI Action:**
 - Configures wide layout (*CODE_PH_0*).
 - Injects CSS design system (*CODE_PH_0*).
 - Renders logo, timestamp, and header card (*CODE_PH_0*).
 - Renders top hero positioning banner (*CODE_PH_0*).

Step 2: Tab Navigation & Filter Initialization

- **Action:** User views the horizontal tab bar. *CODE_PH_0* (*CODE_PH_1*) is selected.
- **Code Execution:**
 - *CODE_PH_0* line 610 triggers *CODE_PH_1*.
 - Calls *CODE_PH_0*.

- **Backend Execution:**
 - FastAPI route **CODE_PH_0** receives GET request.
 - Calls **CODE_PH_0**.
 - Ingests **CODE_PH_0** and returns JSON array of commodities (**CODE_PH_1**, **CODE_PH_2**, **CODE_PH_3**, **CODE_PH_4**, **CODE_PH_5**, etc.).
 - **UI Action:** Populates the commodity selectbox dropdown (**CODE_PH_0**).
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Step 3: Predictive Cargo Analysis Execution

- **Action:** User selects **CODE_PH_0** and **CODE_PH_1** horizon.
 - **Code Execution:**
 - **CODE_PH_0** invokes **CODE_PH_1**.
 - **Backend Execution:**
 1. **CODE_PH_0** triggers **CODE_PH_1**.
 2. Filters **CODE_PH_0** DataFrame for **CODE_PH_1**.
 3. Resamples time-series data to continuous monthly intervals (**CODE_PH_0** to **CODE_PH_1**).
 4. Runs Holt-Winters Exponential Smoothing to extract seasonal components and trend vectors.
 5. Fits Ridge Regression on time-lagged features (**CODE_PH_0**, **CODE_PH_1**) to generate 6-month predictions.
 6. Calculates residual standard error to construct upper and lower 95% dynamic prediction interval arrays.
 7. Maps facility metadata: **CODE_PH_0**, **CODE_PH_1**.
 8. Computes WAPE score (**CODE_PH_0**) and accuracy (**CODE_PH_1**).
 9. Calculates driver weights: **CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**.
 10. Returns full JSON dictionary to frontend.
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Step 4: UI Rendering & Visualization

- **Action:** Frontend receives JSON payload.
 - **Code Execution:**
 - Renders **Selected Commodity Analytics Banner** displaying volume, expected growth (**CODE_PH_0**), predictability score (**CODE_PH_1**), and accuracy.
 - Renders **NMPA Facility Banner** displaying berths and draft depth (**CODE_PH_0**).
 - Renders **Plotly Main Forecast Chart**:
 - Trace 1: **CODE_PH_0** (Dark grey line).
 - Trace 2: **CODE_PH_0** (Shaded amber band).
 - Trace 3: **CODE_PH_0** (Dashed gold line).
 - Renders **Explainable Forecast Drivers** cards detailing factor impacts and weights.
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Step 5: Model Validation & Backtesting Trace

- **Action:** User scrolls down to inspect model validation.

- **Code Execution:**
 - **CODE_PH_0** calls **CODE_PH_1**.
 - Backend **CODE_PH_0** performs chronological out-of-sample backtesting against actual test data.
 - Compares **Baseline Model (Seasonal Naive)** vs **Primary ML Model (Ridge Exponential)**.
 - Renders comparison line chart and WAPE/MAPE/MAE/RMSE comparison table.
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Step 6: Interactive What-If Scenario Simulation Trace

- **Action:** User adjusts sliders (**CODE_PH_0**, **CODE_PH_1**, **CODE_PH_2**) and clicks **Run Cargo Scenario Simulation**.
- **Code Execution:**
- **CODE_PH_0** sends POST request **CODE_PH_1** to **CODE_PH_2**.
- Calls **CODE_PH_0**.
- Calculates simulated throughput delta (**CODE_PH_0**), net volume delta (**CODE_PH_1**), and capacity stress level (**CODE_PH_2**).
- Returns scenario series arrays and operational advisories.
- Renders **Baseline vs Simulated Scenario Forecast Chart** and warning boxes.